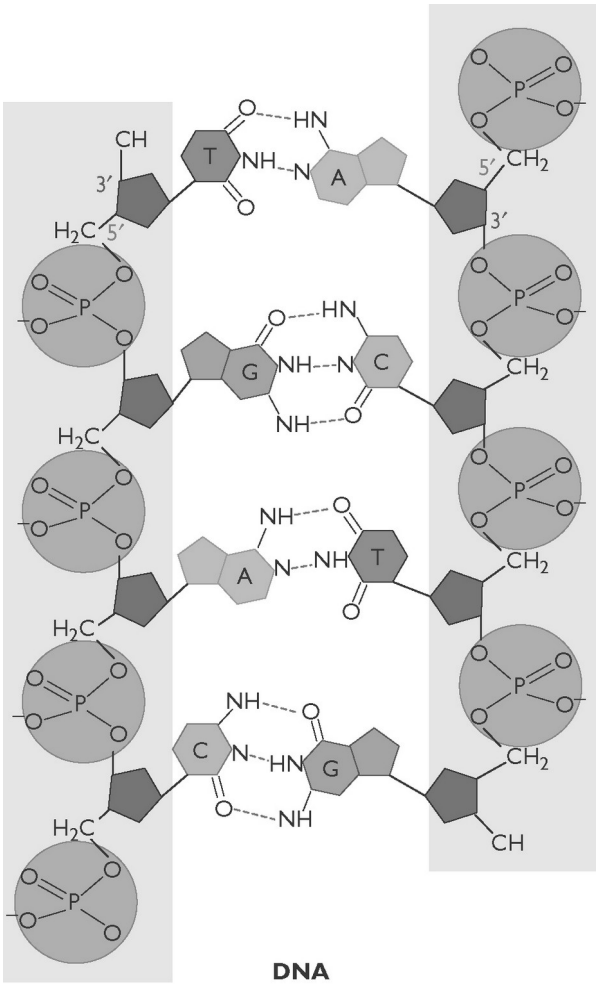
Worksheet 2.2

DNA

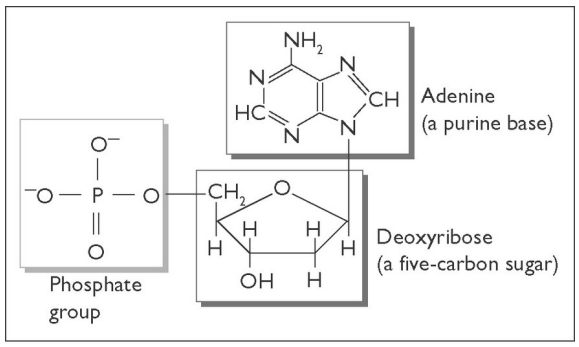
Answers

Nucleotides and DNA

1. The following images show a section of DNA and the composition of a nucleotide.



DNA



Nucleotide

(a) Describe the structure of DNA in terms of nucleotides.

The DNA molecule is composed of 2 chains of 4 different nucleotides.

(b) Identify the molecules in the nucleotides that form the 'backbone' of the DNA 'ladder'.

Deoxyribose sugar and a phosphate group

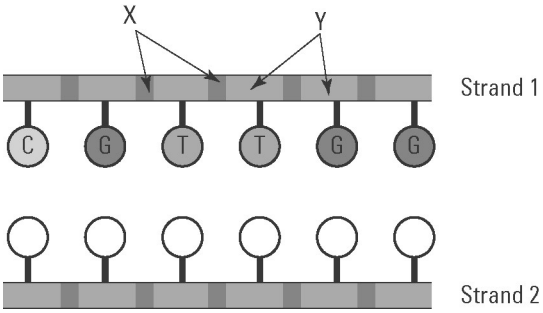
(c) State the pairing arrangement of the four types of nitrogen bases.

A binds to T, and G binds to C.

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Two stranded DNA

2. The following diagram is a simplified section of the double-stranded DNA molecule.



(a) Identify components X and Y.

X: Phosphate

Y: Deoxyribose sugar

(b) In strand 2 on the diagram, record the **letters** of the matching nucleobase (nitrogen base).

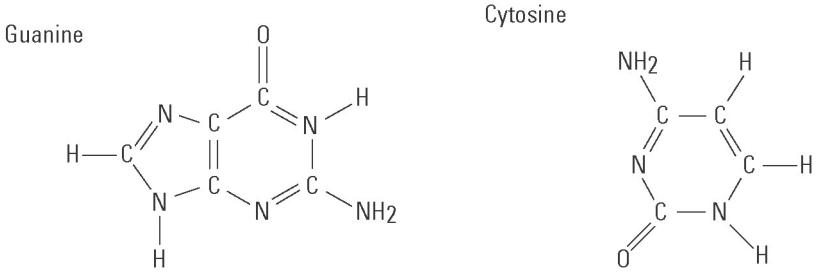
Base order (L to R) = G, C, A, A, C, C

(c) What shape is a DNA molecule?

Double helix

Guanine and cytosine

3. The structures of guanine and cytosine are shown below.



(a) Count the number of C, H, N and O atoms in each molecule.

Guanine:

C: 5 H: 5 N: 5 O: 1

Cytosine:

C: 4 H: 5 N: 3 O: 1

(b) Write the chemical formula for each molecule.

Guanine: $C_5H_5N_5O$

Cytosine: $C_4H_5N_3O$

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